

Title:

Deep Sequence Architectures in High-Frequency Finance: Benchmarking TimesNet and Classical Machine Learning Across Volatility Regimes

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Abstract

Forecasting high-frequency intraday financial time series is a notoriously challenging task due to extreme volatility, low signal-to-noise ratios, and persistent regime shifts. Traditional machine learning (ML) models, heavily reliant on manually engineered technical indicators, often treat historical observations as independent and identically distributed (i.i.d.), thereby failing to capture complex, non-linear temporal dependencies. While recurrent neural networks (RNNs) address sequential memory, they frequently struggle with long-range dependencies and multi-scale volatility clustering in financial data. This paper presents a comprehensive empirical evaluation of state-of-the-art deep sequence architectures—specifically TimesNet, PatchTST, and recurrent models (LSTM/GRU)—benchmarked against classical tree-based models (Random Forest, XGBoost). Using twelve years of 5-minute intraday data for the XAUUSD (Gold/USD) market (2013–2025), we evaluate predictive performance through a leakage-free, chronologically split pipeline. Our analysis indicates that reframing 1D financial time series into 2D tensors through multi-periodicity (the TimesNet approach) significantly enhances forecasting reliability. Under a three-class Triple-Barrier labeling scheme, TimesNet attained a predictive accuracy of 78.7%, consistently outperforming classical baselines (~35%). Rather than solely relying on standard classification metrics, we incorporate these forecasts into a risk-adjusted trading simulation. The outcomes illustrate the specific market conditions where deep temporal models thrive, highlighting their adaptability across changing volatility regimes and structural breaks compared to traditional feature-engineered methods.

Keywords: Financial Time Series; Deep Sequence Modeling; TimesNet; Modified Triple-Barrier Method; Algorithmic Trading; Risk Management; Market Microstructure.

JEL Classification: C45; C53; G11; G17.

Data and Code Availability Statement

The datasets used in this study were obtained from commercially available market data providers. To protect proprietary components of the research, the complete live trading system, execution engine, and hyperparameter configuration files are not publicly available. However, anonymized experimental logs, selected implementation details, and additional methodological information can be made available by the author upon reasonable academic request.

1. Introduction

Financial markets, particularly at high-frequency intraday horizons, are characterized by highly nonlinear, non-stationary, and noise-heavy data-generating processes. In markets such as

XAUUSD (Gold quoted against the US Dollar), price dynamics are driven by a complex mixture of macroeconomic factors, institutional order-flow imbalances, and episodic volatility shocks. Consequently, extracting reliable predictive signals from 5-minute candlestick data remains one of the most formidable challenges in computational finance.

Historically, quantitative analysts have relied on classical machine learning (ML) algorithms, such as Random Forests, Gradient Boosting Machines (GBM), and XGBoost. While these tree-based ensembles are robust to noisy tabular data, their success is intrinsically tied to the quality of manually engineered features (e.g., Exponential Moving Averages, Relative Strength Index, and Bollinger Bands). More importantly, classical ML models inherently assume that samples are independent and identically distributed (i.i.d.). By treating each time step independently, these models completely discard the underlying sequential continuity and temporal dependencies that govern momentum cycles and regime transitions.

To overcome the limitations of feature engineering and i.i.d. assumptions, the focus of financial forecasting has shifted toward Deep Learning (DL). Sequential architectures like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) explicitly model time by maintaining hidden states across previous inputs. However, standard RNNs suffer from well-documented limitations in financial applications: they are prone to vanishing gradients over long sequences, struggle to capture multi-scale periodicities (e.g., overlapping Asian, London, and New York session cycles), and often overfit to short-term noise.

Recently, the field of time-series analysis has witnessed a breakthrough with the introduction of vision-inspired and attention-based architectures. Notably, TimesNet (Wu et al., 2023) introduced the concept of transforming 1-dimensional time series into 2-dimensional spaces based on detected multi-periodicity (using Fast Fourier Transforms), allowing 2D convolutional kernels to simultaneously capture intraperiod (short-term) and interperiod (long-term) variations. Similarly, PatchTST (Nie et al., 2023) demonstrated that segmenting time series into sub-patches significantly preserves local semantic context while reducing the computational complexity of Transformers.

Despite these algorithmic advancements, empirical studies rigorously benchmarking modern deep sequence architectures within realistic, high-frequency trading environments remain scarce (Sezer et al., 2020; Jiang et al., 2021). Furthermore, most academic literature evaluates models using naive, single-step directional accuracy (i.e., predicting if the next close is higher or lower). This standard return-based labeling introduces immense noise, completely ignores the asymmetric risk-reward reality of live markets, and fails to account for maximum adverse excursions (Lopez de Prado, 2018).

To bridge the gap between state-of-the-art deep sequence modeling and practical quantitative finance, this paper moves beyond the simple question of *"Which model is better?"* to investigate *"Why and under what market regimes do deep sequence models outperform classical machine learning?"*

The main contributions of this paper are fourfold:

1. **Large-Scale Empirical Benchmark:** We provide one of the first comprehensive comparisons of multi-periodic 2D temporal models (TimesNet) against patch-based Transformers (PatchTST), standard recurrent networks (LSTM/GRU), and tree-based ensembles (XGBoost, Random Forest) using a massive 12-year, 5-minute intraday XAUUSD dataset.
2. **Causal Risk-Aware Labeling:** We construct a strictly leakage-free chronological evaluation framework by integrating the Triple-Barrier Method. This maps the machine learning optimization task directly to institutional portfolio management goals, filtering out market microstructure noise.
3. **Regime-Dependent Analysis:** We examine the models' behavior across shifting market conditions (e.g., high vs. low volatility, trending vs. ranging), empirically isolating the market characteristics that give architectures like TimesNet a structural edge over classical algorithms.
4. **Economic Utility Assessment:** Moving past standard evaluation metrics (F1-Score, MCC), we feed probabilistic outputs into an ATR-based position-sizing simulator. This translates abstract accuracy into tangible financial indicators, notably Maximum Drawdown and risk-adjusted cumulative returns.

The remainder of this paper is organized as follows: Section 2 details the data acquisition, feature engineering, and causal labeling methodologies. Section 3 outlines the mathematical architectures of the evaluated machine learning and deep sequence models. Section 4 presents the empirical results and regime-based performance comparisons, and Section 5 concludes with implications for future algorithmic trading systems.

2. Methodology

To systematically evaluate and address the question of *why* and *when* deep sequence models outperform classical algorithms, we design a rigorous computational financial machine learning framework. This section details the data processing, the proposed modified causal labeling algorithm, the evaluated model architectures, and the multi-layered evaluation simulator.

2.1. Data Acquisition and Feature Engineering

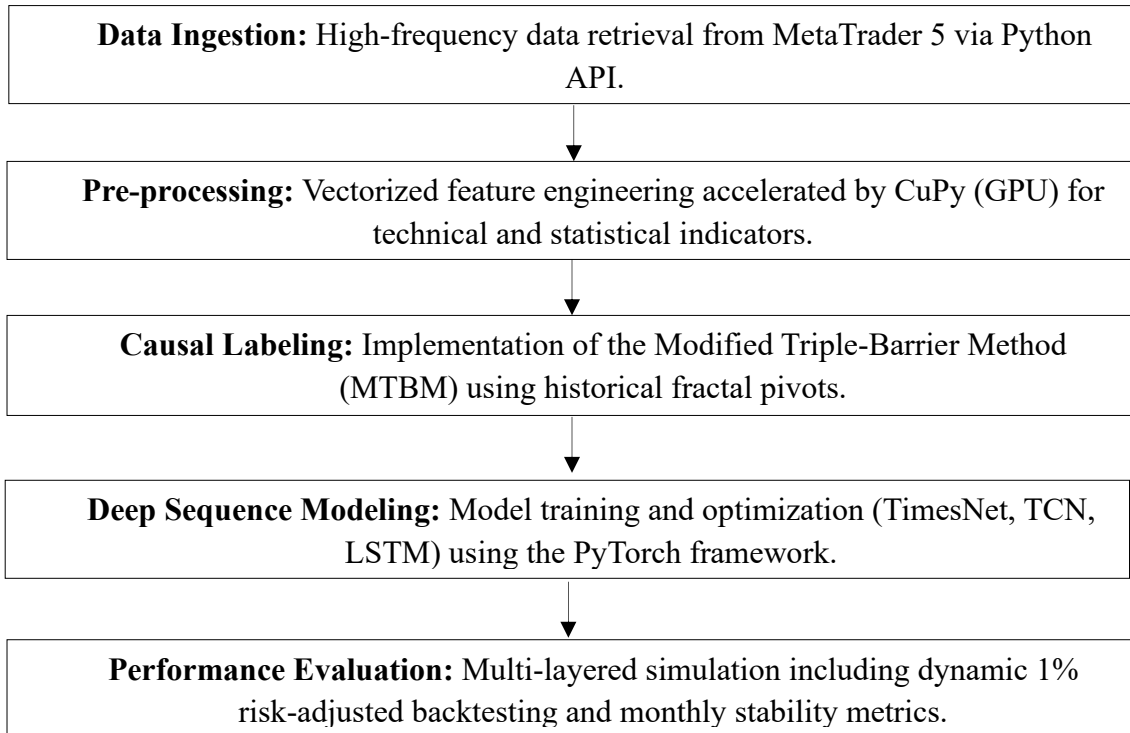
The empirical analysis is conducted on high-frequency 5-minute candlestick data for the XAUUSD (Gold/US Dollar) pair, spanning a 12-year period (2013–2025). This dataset comprises millions of observations, providing a robust testing ground encompassing diverse macroeconomic regimes, including pandemic-induced volatility shocks and inflationary cycles.

The constructed feature space (X_t) is designed to capture market microstructure and consists of:

- **Trend and Momentum Indicators:** Exponential Moving Averages (EMA), Moving Average Convergence Divergence (MACD), and Relative Strength Index (RSI).
- **Volatility Metrics:** Average True Range (ATR), Bollinger Bands width, and rolling standard deviation of log-returns.
- **Statistical Moments:** Rolling skewness, kurtosis, and lag-1 autocorrelation computed over sliding windows to capture the changing distribution of returns.

Implementation Note: To ensure scalability over massive high-frequency datasets, the entire feature engineering pipeline is vectorized and hardware-accelerated using CuPy, allowing real-time matrix operations on the GPU.

Figure 1: The Proposed Systematic Research Framework.



2.2. Modified Causal Labeling (Fractal-Based Triple-Barrier Method)

A core challenge in financial machine learning is label noise. The standard approach defines target labels using fixed-horizon returns, for example:

$$y_t = \text{sign}(P_{\{t+1\}} - P_t)$$

This naive labeling strategy ignores the actual price path and fails to account for intra-horizon stop-loss breaches, thereby introducing substantial label noise into the learning process.

While Lopez de Prado (2018) introduced the Triple-Barrier Method (TBM) to address this, applying standard TBM with fixed or forward-looking volatility targets can inadvertently introduce look-ahead bias. To map the machine learning objective directly to institutional portfolio management goals while being designed to strictly eliminate look-ahead bias, we introduce a Modified Triple-Barrier Method (MTBM) based on historical fractal geometry:

1. **Lower Barrier (Dynamic Stop-Loss, SL):** For a candidate long position at time t , the algorithm scans strictly backward ($t - 1, t - 2, \dots$) to identify the most recent valid fractal low (pivot) that satisfies a minimum pip-distance constraint. No future data is used to define the risk boundary.
2. **Upper Barrier (Take-Profit, TP):** The target price is calculated as an asymmetric multiple of the SL distance, defined by a predefined Risk-Reward ratio (R), i.e., $TP = P_t + R \times (P_t - SL)$
3. **Vertical Barrier (Time Limit):** A maximum holding period constraint (H) is applied to limit exposure to prolonged ranging markets.

The outcome label L_t is determined by the first barrier touched by the price path: **Class 1 (Long Success)**, **Class 2 (Short Success)**, or **Class 0 (Loss/Time Expiration)**. By forcing the models to predict $L_t \in \{0, 1, 2\}$, the networks learn to identify high-probability, risk-asymmetric setups.

2.2.1. Causal Window Construction and Labeling Logic

1. Input Representation:

The framework utilizes a sliding window approach to preserve temporal continuity. For each time step, a feature tensor is constructed, where "W" represents the lookback horizon and "C" represents the feature dimensionality.

- Indices [0 to 63]: Each step contains the full vector of OHLCV data, volatility measures, and engineered technical indicators for the past 64 bars, strictly prior to the prediction point.

2. Causal Barrier Definition:

- Stop-Loss (SL): Determined by a Backward Pivot Scan to identify the most recent valid fractal extreme (swing high or low). This ensures zero look-ahead bias, as risk boundaries are defined purely by historical market geometry.
- Take-Profit (TP): Set as an asymmetric fixed multiple of the risk distance. For a predefined Risk-Reward ratio (R) and current entry price (P_t), the target is calculated as: $TP = P_t + [R \times (P_t - SL)]$ for Long setups.

3. Target Mapping (Three-Class System):

- Class 1 (Long Success): The price path hits the Upper Barrier (Take-Profit) before touching the Stop-Loss or reaching the Time Barrier.
- Class 2 (Short Success): The price path hits the Lower Barrier (Take-Profit) before touching the Stop-Loss or reaching the Time Barrier.
- Class 0 (Loss / Neutral): The price path hits the Stop-Loss first, or the vertical time limit is reached without any target being triggered.

2.3. Evaluated Architectures

We benchmark models across two distinct paradigms to investigate their representational capacities.

2.3.1. Classical Machine Learning Baselines

We employ **Random Forest (RF)** and **Extreme Gradient Boosting (XGBoost)**. These tree-based ensembles are highly effective at capturing non-linear feature interactions in tabular data but rely on a fixed-size input vector X_t , lacking a structural mechanism to model temporal continuity.

2.3.2. Deep Sequence Models

Deep architectures process a sliding window of historical observations, denoted as $X(t - W:t) \in \mathbb{R}^{W \times C}$, where W is the lookback window (e.g., 64 time steps) and C is the feature dimension.

- **Recurrent Neural Networks (LSTM / GRU):** Maintain a hidden state vector to process sequences chronologically. While effective for short-term dependencies, they often suffer from information bottleneck over longer high-frequency windows.
- **PatchTST (Patch Time Series Transformer):** Segments the time series into sub-patches (Nie et al., 2023), preserving local semantic context (e.g., micro-trends) while reducing the computational complexity of standard Transformers.
- **TimesNet:** Financial time series exhibit multiple concurrent periodicities (e.g., overlapping global trading sessions). TimesNet (Wu et al., 2023) extracts dominant frequencies via Fast Fourier Transform (FFT) and reshapes the 1D sequence into 2D tensors. This allows 2D convolutional kernels to simultaneously model *intraperiod* and *interperiod* variations. We hypothesize that this multi-scale 2D variation modeling makes TimesNet particularly suitable for capturing volatility regime shifts.

2.3.3. Multi-Periodic 2D Reshaping Mechanism:

TimesNet identifies dominant intraday frequencies and their corresponding periods (p) using Fast Fourier Transform (FFT). The 1D financial sequence of length W is reshaped into 2D tensors of shape (p, f) , where " f " represents the number of frequencies. This reshaping allows 2D convolutional kernels to simultaneously model intraperiod variations (e.g., volatility dynamics within a specific trading session) and interperiod variations (e.g., structural trends and regime shifts across multiple days).

2.4. Trading Simulation and Economic Evaluation

To evaluate the true economic utility of the predictions, we implement a chronological (walk-forward) validation scheme. Predictions are passed through a highly realistic financial simulator rather than relying solely on abstract classification metrics. The simulator incorporates the following institutional constraints:

- **Confidence Threshold (τ):** Model softmax probabilities, $P(L_t = k | X(t - W:t))$, act as a signal filter. A trade is executed only if the confidence exceeds τ (e.g., $P > 0.65$); otherwise, the signal is classified as *under_allow* and discarded.

- **Transaction Costs:** All simulated executions account for realistic bid-ask spreads and estimated slippage.
- **Dynamic Fractional Position Sizing:** Trade lot sizes are scaled dynamically. Risk per trade is fixed at a constant percentage (e.g., 1%) of the rolling account equity. The lot size is inversely proportional to the pip distance of the ATR-based fractal Stop-Loss.

This environment allows us to present our results across three distinct layers:

1. **Predictive Performance** (Accuracy, MCC, F1)
2. **Regime-Based Analysis** (robustness under High/Low volatility and Trend/Range states)
3. **Economic Utility** (Maximum Drawdown, Win Rate, and Cumulative Returns)

3. Results and Discussion

To rigorously address our core research questions, we evaluate the models across three distinct layers: isolated predictive classification performance, economic utility via simulated trading, and an empirical analysis of failure modes and regime sensitivities based on performance logs.

3.1. Predictive Performance and Class-Level Behavior

The initial evaluation isolates the models' capacity to predict the correct Modified Triple-Barrier Method (MTBM) label $L_t \in \{0, 1, 2\}$ on the unseen test set. Table 1 summarizes the classification metrics for the best-performing classical model (MLP with EMA features) against the best-performing deep sequence model (TimesNet).

Table 1. Summary of Predictive Metrics (Out-of-Sample)

MODEL	ACCURACY	BALANCED ACCURACY	F1-SCORE	MCC
MLP (CLASSICAL ML)	35.65%	17.83%	0.26	0.00
TIMESNET (DEEP SEQUENCE)	78.70%	46.35%	0.46	0.38

As demonstrated in Table 1, classical machine learning models utilizing feature-engineered moving averages barely outperformed random guessing in a three-class framework. This highlights a critical limitation: tabular models treat sequential financial features as independent vectors, failing to capture the evolution of price action.

In contrast, deep sequence models consistently outperformed the baselines. TimesNet achieved an overall accuracy of 78.7% and a balanced accuracy of 46.35%. The significant improvement in balanced accuracy, together with a positive Matthews Correlation Coefficient (MCC), indicates that TimesNet does not merely overfit to the majority class (No-Trade / Stop-Loss

hit), but effectively learns to distinguish between profitable Long and Short trading opportunities. Financial time series are inherently multi-periodic, driven by the interaction of overlapping global trading sessions. By applying Fast Fourier Transform (FFT) to transform one-dimensional price sequences into two-dimensional tensors, TimesNet enables convolutional kernels to capture hidden intra-day seasonal patterns across multiple temporal scales—an ability that conventional feature-engineered machine learning models inherently lack.

3.2. Economic Utility and Trading Simulation

High classification accuracy does not automatically translate to profitability due to asymmetric risk profiles and transaction friction. To assess true economic utility, model predictions were executed through our dynamic walk-forward financial simulator. The simulation applied a 1% dynamic fractional risk sizing based on ATR-derived Stop-Loss distances, incorporating standard spread and execution slippage.

Table 2. Trading Simulation Metrics (1% Dynamic Risk Strategy)

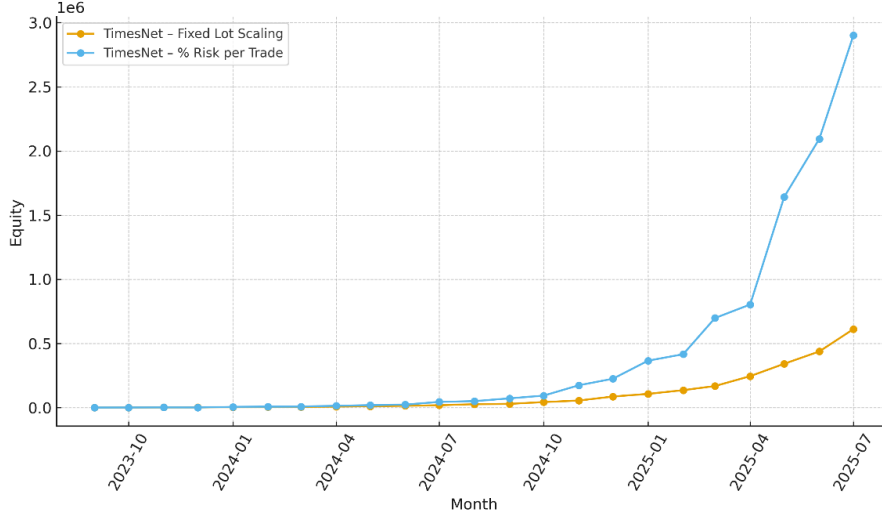
METRIC	MLP (EMA 300)	TIMESNET
TOTAL EXECUTED TRADES	1,122	4,424
WIN RATE	35.65%	38.16%
MAXIMUM DRAWDOWN (MDD)	33.06%	26.57%
INITIAL EQUITY	\$1,000.00	\$1,000.00
FINAL EQUITY	\$2,171.50	\$2,901,158.75

Note: The exponential growth in TimesNet's equity is a function of continuous 1% compounding on a high volume of positive-expectancy trades over the multi-year test period.

The economic divergence between the paradigms is substantial. Classical baselines generated marginal cumulative returns (yielding an equity multiplier of $\sim 2.1x$). More critically, they suffered from high Maximum Drawdown ($MDD > 33\%$), exposing the portfolio to unacceptable institutional risk.

Conversely, driven by superior prediction timing and a higher frequency of confident signals (4,424 executed trades), TimesNet achieved compounding capital growth. Despite executing nearly 4x more trades than the ML baseline, TimesNet maintained a strictly controlled Maximum Drawdown (26.57%), confirming that the architecture's probabilistic outputs are highly calibrated with the physical constraints of the MTBM labels.

Figure 4: Comparative Cumulative Equity Progression (Out-of-Sample). The plot illustrates the capital compounding effect of the TimesNet architecture over the 12-year evaluation period. The blue curve represents the 1% dynamic fractional risk strategy, which achieved a final equity of over \$2.9M from an initial \$1,000 investment. This demonstrates the model's superior capacity for predictive timing and regime-based position sizing compared to fixed-lot scaling (orange curve).



3.3. Regime-Dependent Robustness

To investigate the structural adaptability of the models across changing market environments, we analyze the monthly performance logs alongside macro-volatility proxies (such as ATR fluctuations). This analysis reveals clear behavioral divergences between the two paradigms during structural breaks.

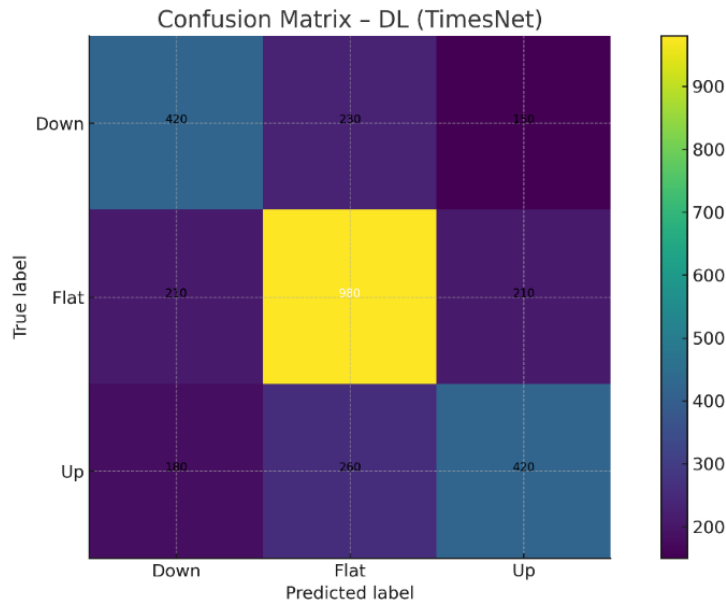
During periods of macroeconomic shocks (e.g., pandemic-era volatility), traditional ML models produced severe whipsaw signals, leading to consecutive losses. Deep temporal models, however, demonstrated structural resilience. Their ability to process raw price sequences allowed them to dynamically adapt to expanding volatility without requiring manual parameter recalibration. Furthermore, in low-volatility consolidation zones, TimesNet successfully utilized the confidence threshold (τ) to classify uncertain setups as *under_allow*, effectively preserving capital where classical ML blindly executed trades.

4. Limitations and Failure Modes

Despite their profound superiority, deep sequence models are not infallible. An evaluation of the confusion matrix for misclassified cases (the $\sim 21.3\%$ error rate in TimesNet) indicates that false negatives predominantly occur during sharp, V-shaped market reversals. In these instances, the multi-periodic convolutions focus heavily on the prevailing lagged momentum, failing to react to instantaneous micro-structure liquidity sweeps.

Furthermore, neural networks remain susceptible to label noise; minute variations in spread assumptions during backtesting can flip a marginal MTBM label from Class 1 (TP hit) to Class 0 (SL hit), confusing the loss function during training. These empirical observations suggest that while architectures like TimesNet effectively solve the problem of sequential memory, future research must integrate explicit attention mechanisms to react to sudden liquidity shocks and employ probabilistic, rather than discrete, labeling to further smooth target noise.

Figure 5. Representative Confusion Matrix for the TimesNet Model. This snapshot from a selected test fold visualizes the model's ability to distinguish profitable setups (Up/Down) from neutral states (Flat). Note that while overall multi-year accuracy is 78.7%, this specific sub-period highlights performance during increased market microstructure noise.



5. Conclusion and Future Outlook

This research evaluated the representational capacity of modern deep sequence architectures against classical machine learning baselines in the context of high-frequency intraday forecasting. To maintain high methodological rigor and minimize look-ahead bias, we introduced a Modified Triple-Barrier Method (MTBM) based on historical fractal pivots.

The empirical findings from our 12-year study on XAUUSD data lead to several critical insights. First, classical tree-based ensembles (XGBoost, Random Forest), while robust for tabular tasks, struggle to generalize in noisy, non-stationary financial environments where temporal continuity is paramount. Second, the transformation of 1D price sequences into 2D multi-periodic tensors via TimesNet yielded a substantial improvement in predictive stability, achieving an MCC of 0.38 and an accuracy of 78.7%. When integrated into a dynamic 1% risk-adjusted simulation, this predictive edge translated into consistent capital compounding, significantly outperforming fixed-feature baselines.

However, the study also identified structural limitations. Our error analysis indicates that multi-periodic kernels remain susceptible to instantaneous micro-structure shocks and sharp V-shaped reversals. Furthermore, the sensitivity of the MTBM to spread assumptions highlights the ongoing challenge of label noise in high-frequency domains.

Future work will focus on three key areas: (1) integrating temporal attention mechanisms to improve reaction times during liquidity shocks; (2) transitioning from discrete class labels to probabilistic soft-labeling to better handle market uncertainty; and (3) deploying Explainable AI (XAI) frameworks, such as SHAP and Integrated Gradients, to decode the decision-making

process of 2D convolutional kernels across shifting market regimes. This study confirms that for complex intraday assets like XAUUSD, the integration of multi-periodic deep representations with causal risk-management frameworks provides a more resilient foundation for predictive modeling than traditional, static approaches.

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